

Comp9417 Week2

Regression

Bingchen Xie

Disclaimer:

These slides are my own supplementary resource and are not official course materials. Any mistakes are my own.

Please refer to the solutions posted on **Moodle** as the official source.

Break Ice

- ❖ Group Assignment
- ❖ Please talk to the people around you
 - ❖ Name
 - ❖ Strengths you can bring to a group, such as coding, maths, writing, testing
 - ❖

Question 1

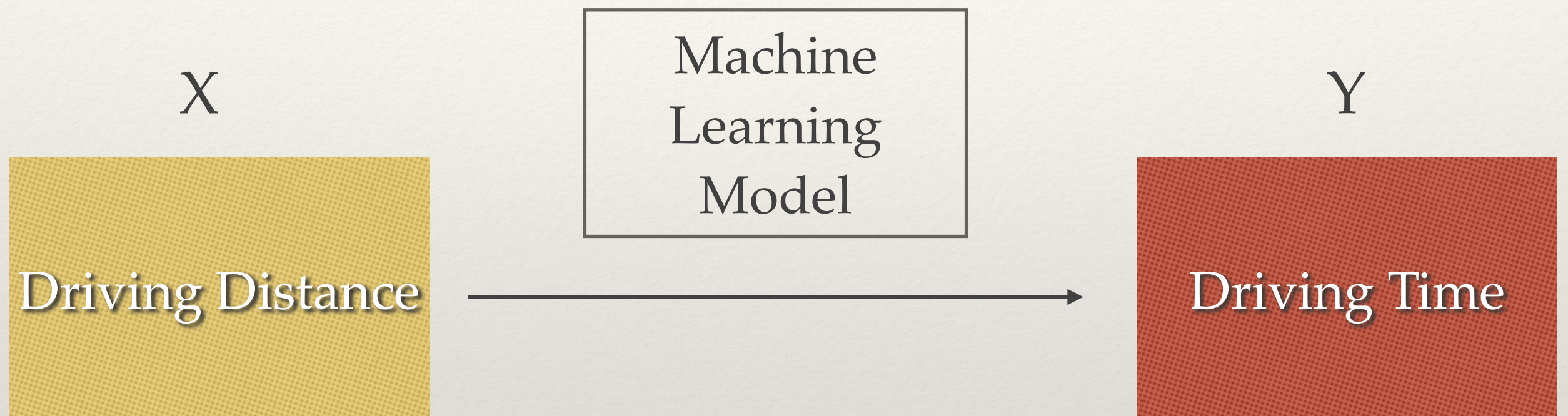
Univariate Least Squares

A *univariate linear regression model* is a linear equation $y = w_0 + w_1x$. Learning such a model requires fitting it to a sample of training data, $(x_1, y_1), \dots, (x_n, y_n)$, so as to minimize the loss (usually Mean Squared Error (MSE)), $\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2$. To find the best parameters w_0 and w_1 that minimize this error function we need to find the error *gradients* $\frac{\partial \mathcal{L}}{\partial w_0}$ and $\frac{\partial \mathcal{L}}{\partial w_1}$. So we need to derive these expressions by taking partial derivatives, set them to zero, and solve for w_0 and w_1 .

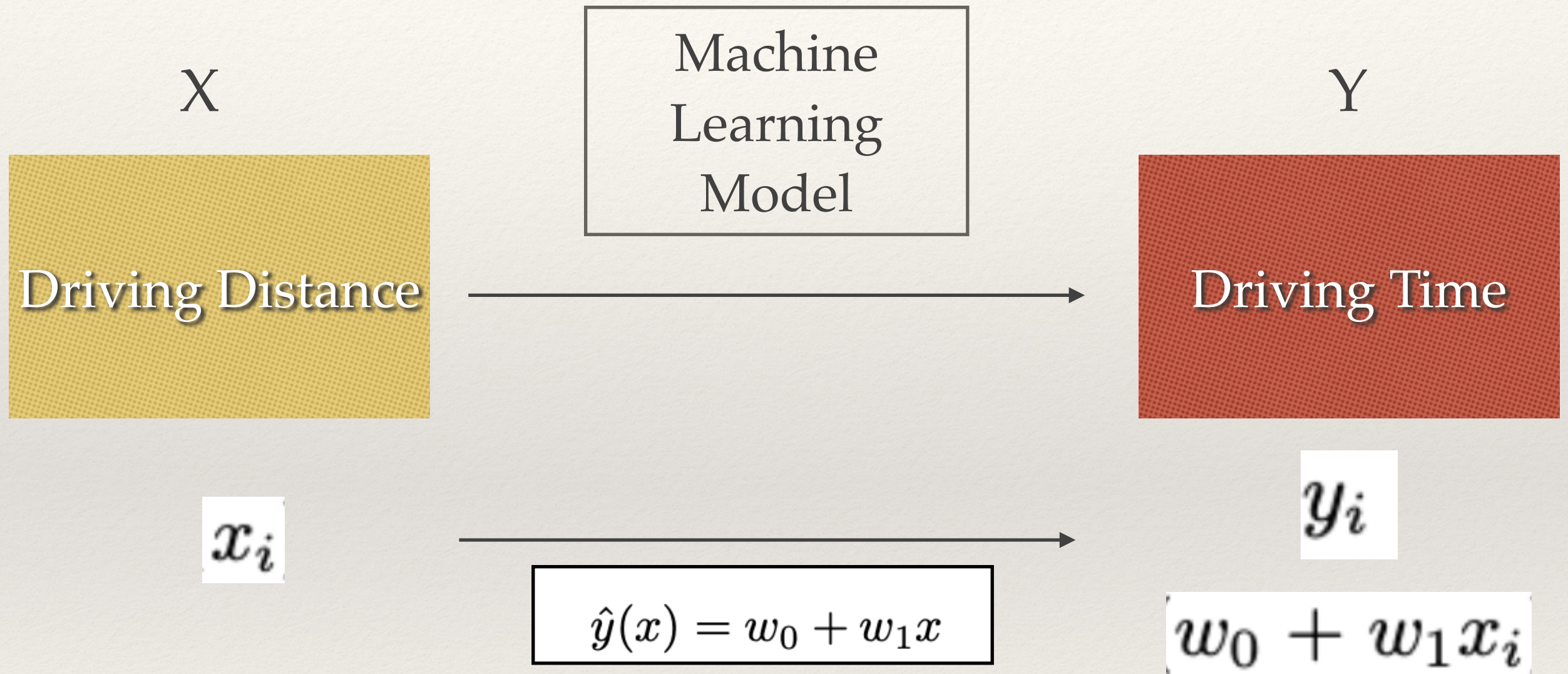
- (a) Derive the least-squares estimates (minimizers of the MSE loss function) for the univariate linear regression model.
- (b) Show that the centroid of the data, i.e. the point $(\bar{x}, \bar{y})$ is always on the least squares regression line.
- (c) To make sure you understand the process, try to solve the following loss function for linear regression with a version of “L2” regularization, in which we add a penalty that penalizes the size of w_1 . Let $\lambda > 0$ and consider the regularised loss

$$\mathcal{L}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1x_i))^2 + \lambda w_1^2$$

Q1 Recap: Regression



Q1 Recap: MSE



Q1 Recap: MSE

x_i



y_i

$w_0 + w_1 x_i$

Error:

$$y_i - (w_0 + w_1 x_i)$$

Squared Error:

$$(y_i - (w_0 + w_1 x_i))^2$$

Mean Squared Error:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

MAE:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n |y_i - (w_0 + w_1 x_i)|$$

Q1a Analyze

- (a) Derive the least-squares estimates (minimizers of the MSE loss function) for the univariate linear regression model.

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

Find the best parameters (w_0 , w_1) to minimize the MSE.

partial derivative of $\mathcal{L}$ with respect to w_0 :

partial derivative of $\mathcal{L}$ with respect to w_1 :

→ Equate them to Zero

Q1a Calculation

partial derivative of $\mathcal{L}$ with respect to w_0 :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w_0} &= \frac{\partial}{\partial w_0} \frac{1}{n} \sum_{i=1}^n (y_i - w_0 - w_1 x_i)^2 \\&= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial w_0} (y_i - w_0 - w_1 x_i)^2 \\&= \frac{1}{n} \sum_{i=1}^n -2(y_i - w_0 - w_1 x_i) \\&= -2 \left[\frac{1}{n} \sum_{i=1}^n y_i - w_0 \frac{1}{n} \sum_{i=1}^n 1 - w_1 \frac{1}{n} \sum_{i=1}^n x_i \right] \\&= -2 [\bar{y} - w_0 - w_1 \bar{x}],\end{aligned}$$

Q1a Calculation

partial derivative of $\mathcal{L}$ with respect to w_1 :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w_1} &= \frac{\partial}{\partial w_1} \frac{1}{n} \sum_{i=1}^n (y_i - w_0 - w_1 x_i)^2 \\ &= \frac{1}{n} \sum_{i=1}^n -2x_i (y_i - w_0 - w_1 x_i) \\ &= -2 \left[\frac{1}{n} \sum_{i=1}^n x_i y_i - w_0 \frac{1}{n} \sum_{i=1}^n x_i - w_1 \frac{1}{n} \sum_{i=1}^n x_i^2 \right] \\ &= -2 \left[\overline{xy} - w_0 \bar{x} - w_1 \overline{x^2} \right],\end{aligned}$$

Q1a Calculation

Equate them to Zero

$$-2 [\bar{y} - w_0 - w_1 \bar{x}] = 0 \implies w_0 = \bar{y} - w_1 \bar{x}$$

$$-2 [\overline{xy} - w_0 \bar{x} - w_1 \overline{x^2}] = 0 \implies w_1 = \frac{\overline{xy} - w_0 \bar{x}}{\overline{x^2}}$$

Q1a Answer

$$w_0 = \bar{y} - w_1 \bar{x} = \left(\bar{y} - \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \bar{x} \right)$$

$$w_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}$$

Q1b Analyze

(b) Show that the centroid of the data, i.e. the point $(\bar{x}, \bar{y})$ is always on the least squares regression line.

$$\begin{aligned}\hat{y}(x) &= w_0 + w_1 x \\ &= \left(\bar{y} - \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \bar{x} \right) + \left(\frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \right) x,\end{aligned}$$

Q1b Analyze

(b) Show that the centroid of the data, i.e. the point $(\bar{x}, \bar{y})$ is always on the least squares regression line.

$$\begin{aligned}\hat{y}(x) &= w_0 + w_1 x \\ &= \left(\bar{y} - \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \bar{x} \right) + \left(\frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \right) x,\end{aligned}$$

$$\hat{y}(\bar{x}) = \bar{y}$$

Q1c Analyze

- (c) To make sure you understand the process, try to solve the following loss function for linear regression with a version of “ $L2$ ” regularization, in which we add a penalty that penalizes the size of w_1 . Let $\lambda > 0$ and consider the regularised loss

$$\mathcal{L}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 + \lambda w_1^2$$

Find the best parameters (w_0, w_1) to minimize the Loss Func.

partial derivative of $\mathcal{L}$ with respect to w_0 :

partial derivative of $\mathcal{L}$ with respect to w_1 :

→ Equate them to Zero

$$\hat{y}(x) = w_0 + w_1 x$$

Q1 Analyze

- (c) To make sure you understand the process, try to solve the following loss function for linear regression with a version of “ $L2$ ” regularization, in which we add a penalty that penalizes the size of w_1 . Let $\lambda > 0$ and consider the regularised loss

$$\mathcal{L}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 + \lambda w_1^2$$

(a)

$$w_0 = \bar{y} - w_1 \bar{x},$$

$$w_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}$$

(c)

$$w_0 = \bar{y} - w_1 \bar{x},$$

$$w_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2 + \lambda}.$$

Question 2

Multivariate Least Squares

Question 2. (Multivariate Least Squares)

In the previous question, we found the least squares solution for the univariate (single feature) problem. We now generalise this for the case when we have p features. Let $x_1, x_2, \dots, x_n$ be n feature vectors (e.g. corresponding to n instances) in $\mathbb{R}^p$, that is:

$$x_i = \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{ip-1} \end{bmatrix}$$

We stack these feature vectors into a single matrix, $X \in \mathbb{R}^{n \times p}$, called the *design* matrix. The convention is to stack the feature vectors so that each row of X corresponds to a particular instance, that is:

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1,p-1} \\ x_{20} & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

Q2 Recap: Regression

Example: The task is to predict house prices based on the available house features

House Size (sq ft)	Num. of Bedrooms	Num. of Bathrooms	Age of House (years)	House Price (\$)
1360	4	2	27	251240
1794	1	2	42	262602
1630	4	2	28	282277
1595	4	1	16	266877
2138	4	1	15	346992
2669	3	2	47	405283
966	2	2	44	143916
1738	1	1	3	278097
830	2	1	37	113612
1982	4	1	7	342283

Q2 Recap: Regression

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1,p-1} \\ x_{20} & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

House Size (sq ft)	Num. of Bedrooms	Num. of Bathrooms	Age of House (years)
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2138	4	1	15
2669	3	2	47
966	2	2	44
1738	1	1	3
830	2	1	37
1982	4	1	7

Q2 Recap: Regression

where the superscript T denotes the transpose operation. Note that it is standard to take the first element of the feature vectors to be 1 to account for the bias term, so we will assume $x_{i0} = 1$ for all $i = 1, \dots, n$. Analogously to the previous question, the goal is to learn a weight vector $w \in \mathbb{R}^p$, and make predictions:

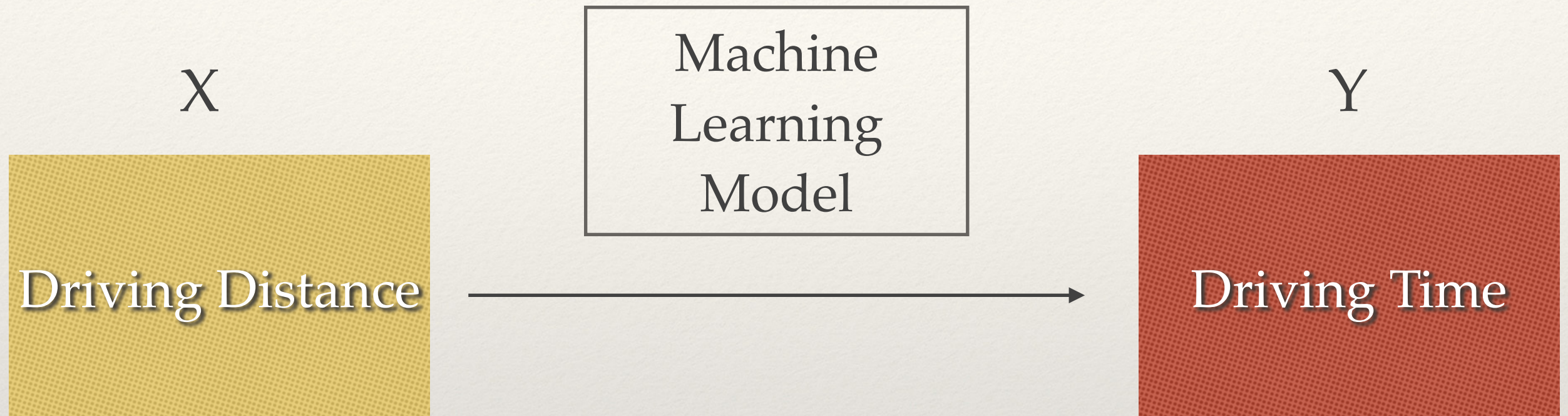
$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_{p-1} x_{i,p-1},$$

$$\begin{aligned} \text{House Price} = & - 8775.58 \\ & + 147.12 \times \text{House Size (sq ft)} \\ & + 9660.37 \times \text{Number of Bedrooms} \\ & + 25691.99 \times \text{Number of Bathrooms} \\ & - 1285.01 \times \text{Age of House (years)} \end{aligned}$$

Compared with Question 1:

$$\hat{y}(x) = w_0 + w_1 x$$

Q2 Analyze: MSE



$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_{p-1} x_{i,p-1},$$

x_i

y_i

$$w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_{p-1} x_{i,p-1}$$

Q2 Analyze

where the superscript T denotes the transpose operation. Note that it is standard to take the first element of the feature vectors to be 1 to account for the bias term, so we will assume $x_{i0} = 1$ for all $i = 1, \dots, n$. Analogously to the previous question, the goal is to learn a weight vector $w \in \mathbb{R}^p$, and make predictions:

$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_{p-1} x_{i,p-1},$$

where $\hat{y}_i$ denotes the i -th predicted value. To solve for the optimal weights in w , we can use the same procedure as before and use the MSE:

$$\begin{aligned} \mathcal{L}(w_0, w_1, \dots, w_{p-1}) &= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_{p-1} x_{i,p-1}))^2. \end{aligned}$$

One approach to solve this would be to take derivatives with respect to each of the p weights and solve the resulting equations, but this would be **extremely** tedious. The more efficient way to solve this problem is to appeal to matrix notation. We can write the above loss as:

$$\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|_2^2,$$

where $\|\cdot\|_2$ is the Euclidean norm. For the remainder of this question, we will assume that X is a full-rank matrix, which means that we are able to compute the inverse of $X^T X$.

Q2 Recap: Euclidean norm

L2 Norm

≥ 0

$$\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|_2^2,$$

where $\|\cdot\|_2$ is the Euclidean norm. For the remainder of this question, we will assume that X is a full-rank matrix, which means that we are able to compute the inverse of $X^T X$.

$$\mathbf{x} = (x_1, x_2, \dots, x_n),$$

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

$$\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$$

$$\mathbf{y} - \hat{\mathbf{y}},$$

$$\|\mathbf{y} - \hat{\mathbf{y}}\|_2 = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

Q2 Analyze

$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_{p-1} x_{i,p-1},$$

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1,p-1} \\ x_{20} & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

Each Line is an Instance.

Q2 Analyze

$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_{p-1} x_{i,p-1},$$

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1,p-1} \\ x_{20} & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

Each Line is an Instance.

$$\hat{\mathbf{y}}$$

$$=$$

$$Xw$$

Q2a

$$\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|_2^2,$$

(a) Show that $\mathcal{L}(w)$ has a critical point:

$$\hat{w} = (X^T X)^{-1} X^T y,$$

note: critical point here means that $(\frac{d\mathcal{L}(w)}{dw})(\hat{w}) = 0$, i.e. the gradient evaluated at the critical point $\hat{w}$ is zero).

Hint 1: if u is a vector in $\mathbb{R}^n$, and v is a fixed vector in $\mathbb{R}^n$, then $\frac{\partial v^T u}{\partial u} = v$.

Hint 2: if A is a fixed $n \times n$ matrix, and if $f = z^T A z$, then $\frac{\partial u^T A u}{\partial u} = Au + A^T u$.

Q2a Analyze

$$\mathcal{L}(w) = \frac{1}{n} \|y - Xw\|_2^2,$$

$$\hat{y}_i = w^T x_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_{p-1} x_{i,p-1},$$

(a) Show that $\mathcal{L}(w)$ has a critical point:

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Hint 2: if A is a fixed $n \times n$ matrix, and if $f = z^T A z$, then $\frac{\partial u^T A u}{\partial u} = A u + A^T u$.

Find the w vector to minimize the Loss Function.

$$\begin{aligned} \|y - Xw\|_2^2 &= (y - Xw)^T (y - Xw) \\ &= y^T y - 2y^T Xw + w^T X^T Xw. \end{aligned}$$

Q1a Analyze

- (a) Derive the least-squares estimates (minimizers of the MSE loss function) for the univariate linear regression model.

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

Find the best parameters (w_0 , w_1) to minimize the MSE.

partial derivative of $\mathcal{L}$ with respect to w_0 :

partial derivative of $\mathcal{L}$ with respect to w_1 :

→ Equate them to Zero

Q2a Analyze

Find the w vector to minimize the Loss Function.

$$\begin{aligned}\|y - Xw\|_2^2 &= (y - Xw)^T (y - Xw) \\ &= y^T y - 2y^T Xw + w^T X^T Xw.\end{aligned}$$

derivatives with respect to w



set equal to zero

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw)$$

Q2a Calculation

derivatives with respect to w



set equal to zero

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw)$$

Hint 1: if u is a vector in $\mathbb{R}^n$, and v is a fixed vector in $\mathbb{R}^n$, then $\frac{\partial v^T u}{\partial u} = v$.

Hint 2: if A is a fixed $n \times n$ matrix, and if $f = z^T A z$, then $\frac{\partial u^T A u}{\partial u} = A u + A^T u$.

Hint 3:

$$\frac{\partial(a^T w)}{\partial w} = a,$$

Q2a Answer

derivatives with respect to w



set equal to zero

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw)$$

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw) = -2X^T y + 2X^T Xw \stackrel{(\text{set})}{=} 0.$$

Solving for w in the above yields:

$$2X^T Xw = 2X^T y \implies \hat{w} = (X^T X)^{-1} X^T y$$

$$\nabla_w \mathcal{L}(w) = -2X^T y + 2X^T Xw.$$

Q2a Analyze

gradient

derivatives with respect to w

$$\nabla_w \mathcal{L}(w) = \frac{\partial \mathcal{L}(w)}{\partial w}$$

$$\frac{d}{dw} (y^T y - 2y^T Xw + w^T X^T Xw)$$

Q2b

The condition $(\frac{d\mathcal{L}(w)}{dw})(\hat{w}) = 0$ is necessary but not sufficient to show that $\hat{w}$ is a (global) minimizer of $\mathcal{L}$, since this point could be a local minimum or a saddle point. Show that the critical point in part (a) is indeed a global minimizer of $\mathcal{L}$.

Hint 1: $\mathcal{L}(w)$ is a function of $w \in \mathbb{R}^p$, and so its Hessian, H , is the $p \times p$ matrix of second order partial derivatives, that is, the (k, l) -th element of H is

$$H_{kl} = \frac{\partial^2 \mathcal{L}(w)}{\partial w_k \partial w_l}.$$

We will often write $H = \nabla_w^2 \mathcal{L}(w)$, where ∇ is the gradient operator, and ∇^2 means taking the gradient twice. Note that the Hessian plays the role of second derivative for multivariate functions.

Hint 2: a function is convex if its Hessian is positive semi-definite, which means that for any vector u ,

$$u^T H u \geq 0.$$

Note also that this condition means that for any choice of u , the product term will always be non-negative.

Hint 3: Any critical point of a convex function is a global minimum.

Q2b Analyze: Hessian

Gradient

$$\nabla_w \mathcal{L}(w) = \frac{\partial \mathcal{L}(w)}{\partial w}$$

Hessian

$$H = \nabla_w^2 \mathcal{L}(w)$$

$$\nabla_w^2 \mathcal{L}(w) = \nabla_w (\nabla_w \mathcal{L}(w))$$

Hessian plays the role of second derivative for multivariate functions.

Q2b Calculation

take derivative again

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw) = -2X^T y + 2X^T Xw$$

$$\begin{aligned}\nabla_w^2 \mathcal{L}(w) &= \nabla_w(\nabla_w \mathcal{L}(w)) \\ &= \nabla_w(-2X^T y + 2X^T Xw)\end{aligned}$$

Hint:

$$\frac{\partial(a^T w)}{\partial w} = a,$$

Q2b Calculation

take derivative again

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw) = -2X^T y + 2X^T Xw$$

Hint:

$$\frac{\partial(a^T w)}{\partial w} = a,$$

$$\begin{aligned}\nabla_w^2 \mathcal{L}(w) &= \nabla_w(\nabla_w \mathcal{L}(w)) \\ &= \nabla_w(-2X^T y + 2X^T Xw) \\ &= 2X^T X.\end{aligned}$$

Q2b Analyze

$$\begin{aligned}\nabla_w^2 \mathcal{L}(w) &= \nabla_w (\nabla_w \mathcal{L}(w)) \\ &= \nabla_w (-2X^T y + 2X^T X w) \\ &= 2X^T X.\end{aligned}$$

$$u^T (2X^T X) u$$

Hint 2: a function is convex if its Hessian is positive semi-definite, which means that for any vector u ,

$$u^T H u \geq 0.$$

Note also that this condition means that for any choice of u , the product term will always be non-negative.

Hint 3: Any critical point of a convex function is a global minimum.

Hint 4:

L2 Norm

$$\|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$$

$$\geq 0$$

Q2b Analyze

$$u^T (2X^T X) u$$

Hint 2: a function is convex if its Hessian is positive semi-definite, which means that for any vector u ,

$$u^T H u \geq 0.$$

Note also that this condition means that for any choice of u , the product term will always be non-negative.

Hint 3: Any critical point of a convex function is a global minimum.

$$u^T (2X^T X) u = 2(Xu)^T (Xu) = 2\|Xu\|_2^2 \geq 0,$$

since norms are always non-negative. Therefore, $\mathcal{L}$ is indeed convex, and so the critical point found earlier is indeed a global minimizer of $\mathcal{L}$.

Q2c

Question 2. (Multivariate Least Squares)

In the previous question, we found the least squares solution for the univariate (single feature) problem. We now generalise this for the case when we have p features. Let $x_1, x_2, \dots, x_n$ be n feature vectors (e.g. corresponding to n instances) in $\mathbb{R}^p$, that is:

$$x_i = \begin{bmatrix} x_{i0} \\ x_{i1} \\ \vdots \\ x_{ip-1} \end{bmatrix}$$

$$x_i = \begin{bmatrix} 1 \\ x_{i1} \end{bmatrix} \in \mathbb{R}^2,$$

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} = \begin{bmatrix} x_{10} & x_{11} & \cdots & x_{1,p-1} \\ x_{20} & x_{21} & \cdots & x_{2,p-1} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n0} & x_{n1} & \cdots & x_{n,p-1} \end{bmatrix}$$

- (c) In the next parts, we will use the formula derived above to verify our solution in the univariate case. We assume that $p = 2$, so that we have a two dimensional feature vector (one term for the intercept, and another for our feature). Write down the following values:

$$x_i, \quad y, \quad w, \quad X, \quad X^T X, \quad (X^T X)^{-1}, \quad X^T y.$$

Q2c Answer

$$x_i = \begin{bmatrix} 1 \\ x_{i1} \end{bmatrix} \in \mathbb{R}^2, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \in \mathbb{R}^2$$

$$X = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{21} \\ \vdots & \vdots \\ 1 & x_{n1} \end{bmatrix} \in \mathbb{R}^{n \times 2} \quad X^T X = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & n\bar{x}^2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$(X^T X)^{-1} = \frac{1}{n(\bar{x}^2 - \bar{x}^2)} \begin{bmatrix} \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}, \quad X^T y = \begin{bmatrix} n\bar{y} \\ n\bar{x}\bar{y} \end{bmatrix}$$

Q2d

(d) Compute the least-squares estimate for the $p = 2$ case using the results from the previous part.

$$\frac{d}{dw}(y^T y - 2y^T Xw + w^T X^T Xw) = -2X^T y + 2X^T Xw \stackrel{(\text{set})}{=} 0.$$

Solving for w in the above yields:

$$2X^T Xw = 2X^T y \implies \hat{w} = (X^T X)^{-1} X^T y$$

Q2d

(d) Compute the least-squares estimate for the $p = 2$ case using the results from the previous part.

$$\hat{w} = (X^T X)^{-1} X^T y$$

$$x_i = \begin{bmatrix} 1 \\ x_{i1} \end{bmatrix} \in \mathbb{R}^2, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \quad w = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \in \mathbb{R}^2$$

$$X = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{21} \\ \vdots & \vdots \\ 1 & x_{n1} \end{bmatrix} \in \mathbb{R}^{n \times 2} \quad X^T X = \begin{bmatrix} n & n\bar{x} \\ n\bar{x} & n\bar{x}^2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$(X^T X)^{-1} = \frac{1}{n(\bar{x}^2 - \bar{x}^2)} \begin{bmatrix} \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}, \quad X^T y = \begin{bmatrix} n\bar{y} \\ n\bar{x}\bar{y} \end{bmatrix}$$

Q2d Answer

$$w_0 = \bar{y} - w_1 \bar{x} = \left(\bar{y} - \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} \bar{x} \right)$$

$$w_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}$$

Q2e

- (e) Consider the following problem: we have inputs $x_1, \dots, x_5 = 3, 6, 7, 8, 11$ and outputs $y_1, \dots, y_5 = 13, 8, 11, 2, 6$. Compute the least-squares solution and plot the results both by hand and using python. Finally, use the sklearn implementation to check your results.

$$w_1 = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}$$

$$w_0 = \bar{y} - w_1 \bar{x}$$

Q2e Coding

Import

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn import linear_model
```

Training Data

```
# Training data
y = np.array([13, 8, 11, 2, 6])
x = np.array([3, 6, 7, 8, 11])
n = x.shape[0] #number of instances
```

Q2e Coding

Fit the model

```
print(x)
x = x.reshape(-1,1)
print(x)
# Fit the model using scikit-learn
model = linear_model.LinearRegression()
model.fit(x, y)

print("Scikit-learn coefficients:")
print(model.intercept_, model.coef_[0])
```

```
[ 3  6  7  8 11]
[[ 3]
 [ 6]
 [ 7]
 [ 8]
[11]]
```

```
Scikit-learn coefficients:
15.0 -1.0
```

Q2e Coding

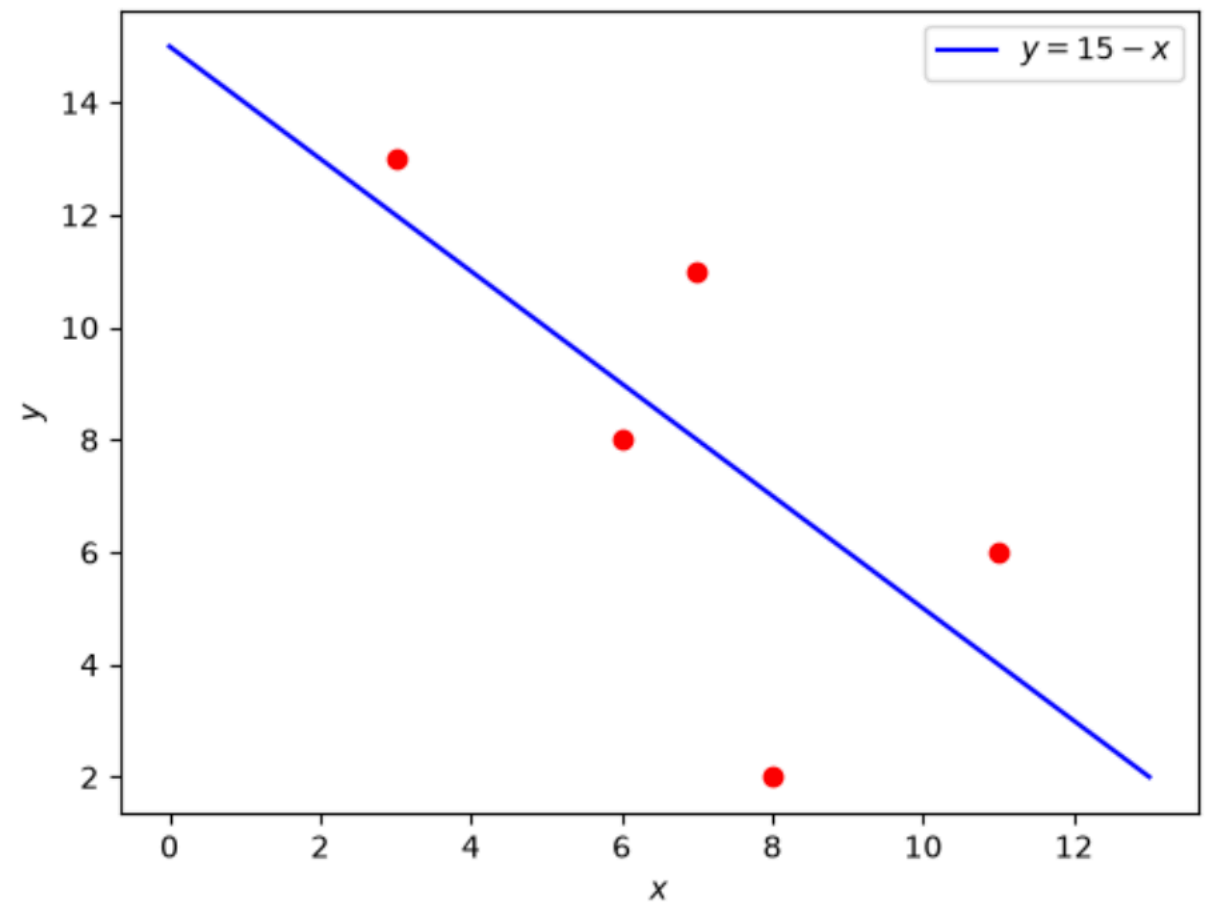
Plot the data

```
# Plot the data points and the fitted regression line
xx = np.linspace(start: 0, stop: 13, num: 1000)

plt.scatter(x, y, color="red")
plt.plot(
    *args: xx,
    15 - xx,
    color="blue",
    label=r"$y = 15 - x$"
)

plt.legend()
plt.xlabel(r"$x$")
plt.ylabel(r"$y$")

plt.savefig("LSLine.png")
plt.show()
```



Q2g

(g) Discuss the idea of a *feature map*. How would you use a feature map in the context of least squares regression?

a feature map is any function $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^K$

$p=1$

$$\phi : \mathbb{R} \rightarrow \mathbb{R}^4, \quad \phi(x) = [x, x^2, x^3, \log x]^T,$$

$p=3$

$$x = [x_1, x_2, x_3]^T,$$

$$\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \phi(x) = \left[\frac{1}{2}(x_1 + x_2), x_3^2 \right]^T.$$

Num. of Bedrooms	Num. of Bathrooms	Age of House (years)
4	2	27
1	2	42
4	2	28
4	1	16
4	1	15
3	2	47

Q2h

(h) The mean-squared error (MSE) is

$$\text{MSE}(w) = \frac{1}{n} \|y - Xw\|_2^2,$$

whereas the sum of squared errors (SSE) (also referred to as the residual sum of squares (RSS)) is

$$\text{SSE}(w) = \|y - Xw\|_2^2.$$

Are the following statements True or False. Explain why.

(i) $\arg \min_{w \in \mathbb{R}^p} \text{MSE}(w) = \arg \min_{w \in \mathbb{R}^p} \text{SSE}(w)$

(ii) $\min_{w \in \mathbb{R}^p} \text{MSE}(w) = \min_{w \in \mathbb{R}^p} \text{SSE}(w)$

Notation: recall that $\min_x g(x)$ is the smallest value of $g(x)$, whereas $\arg \min_x g(x)$ is the value of x that minimizes $g(x)$. So $\min_x (x - 2)^2 = 0$ but $\arg \min_x (x - 2)^2 = 2$.

$$\text{MSE}(w) = \text{SSE}(w) / n$$

Question 3

Question 3. (Population Versus Sample Parameters)

- (a) What is the difference between a population and a sample?
- (b) What is a population parameter? How can we estimate it?

Q3 Analyze

Question 3. (Population Versus Sample Parameters)

- (a) What is the difference between a population and a sample?
- (b) What is a population parameter? How can we estimate it?

Sample $\longrightarrow$ Inference about Population

Population	Sample
Population mean μ	Sample mean $\bar{x}$
Population variance σ^2	Sample variance s^2
Parameter	Statistic
Usually unknown	Can be calculated from the sample data